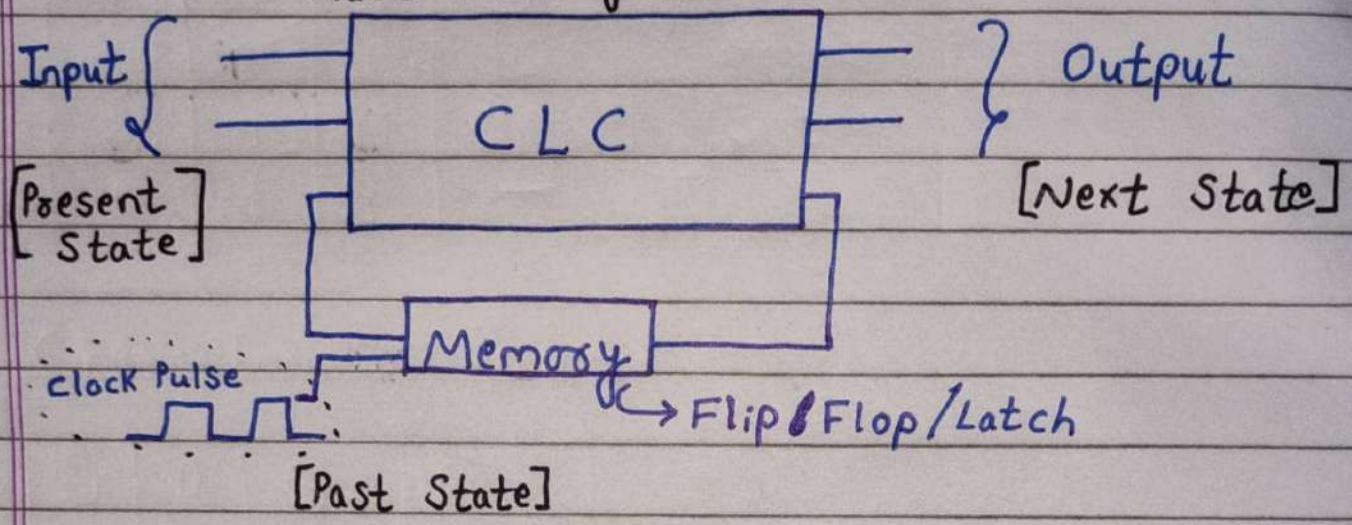
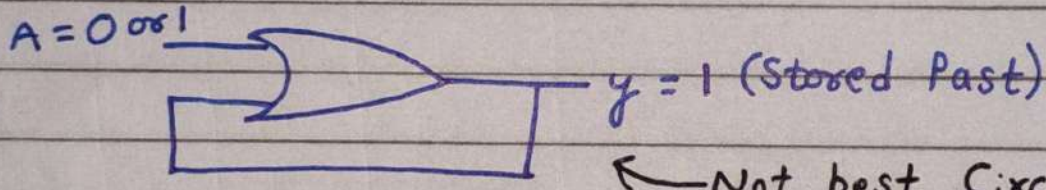
Module-III

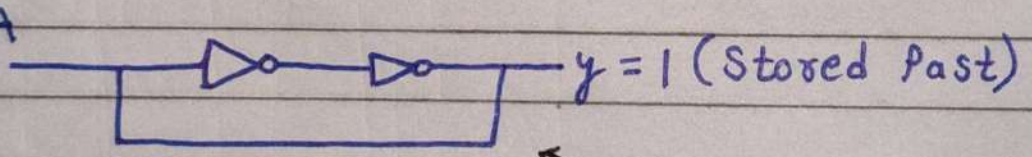
Sequential Logic Circuit

↳ Have Memory for Past Data





← Not best Circuit Here.



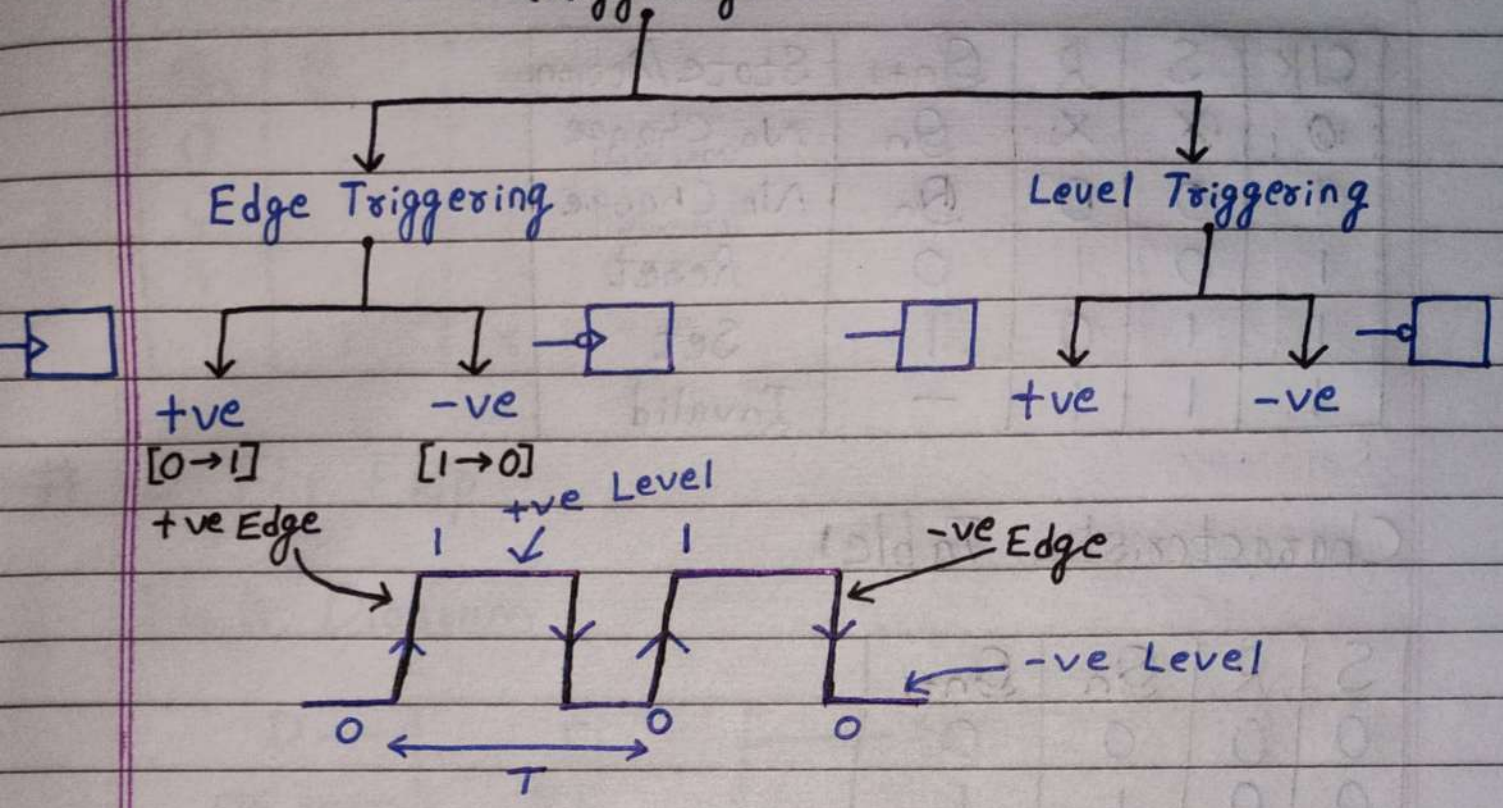
← Used similar for flip-flop

Types:

1. Synchronous Sequential Logic
   ↳ Clock Pulse, Consume Higher Energy

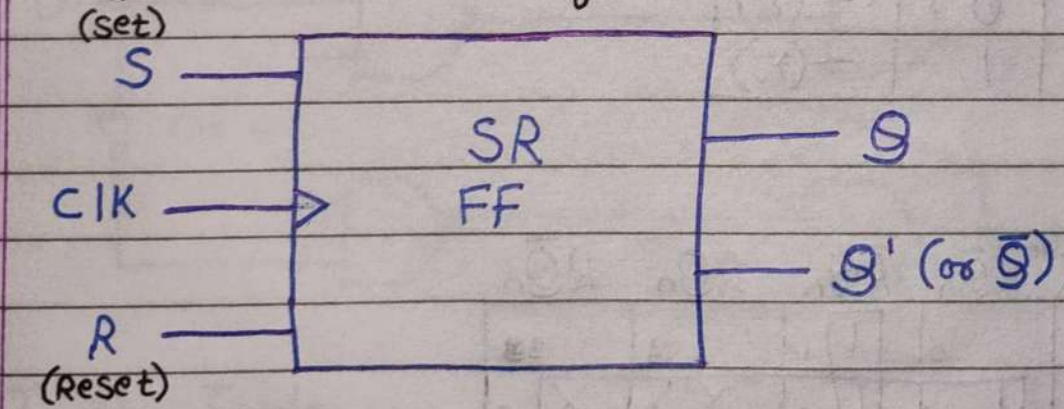
2. Asynchronous Sequential Logic
   ↳ No Clock Pulse, Consume Lower Energy

Clock Pulse
(Triggering)

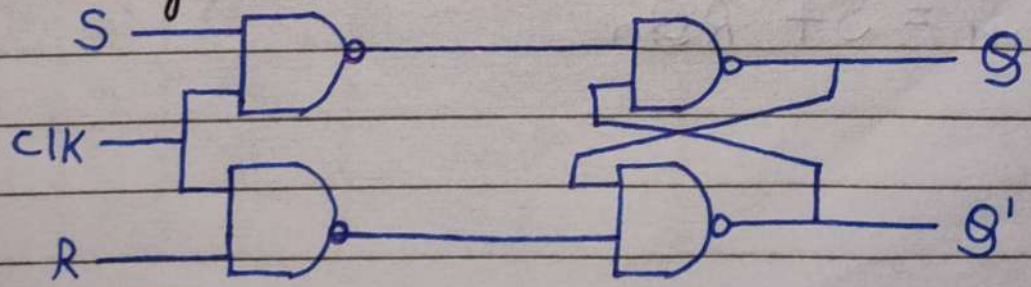


S-R Flip Flop

Block Diagram



Logic Diagram



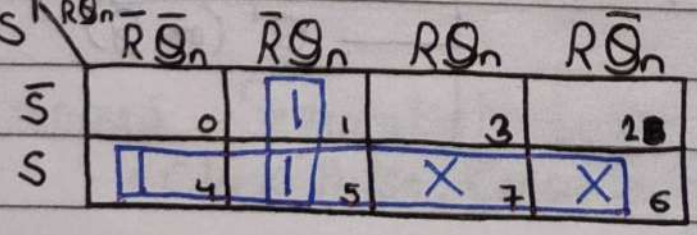
Truth Table:

CIK	S	R	Q_{n+1}	State/Action
0	X	X	Q_n	No Change (Memory)
1	0	0	Q_n	No Change (Memory)
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	–	Invalid

Characteristic Table:

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	–(X)
1	1	1	–(X)

K-map



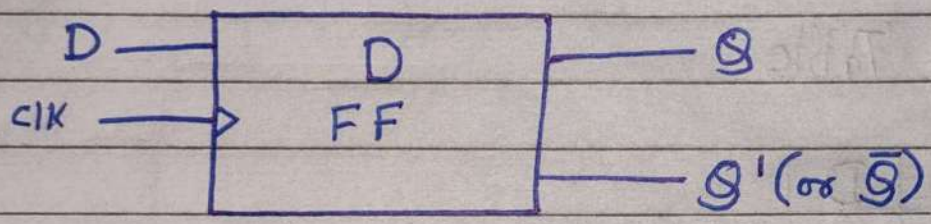
$$Q_{n+1} = S + \bar{R}Q_n$$

Excitation Table

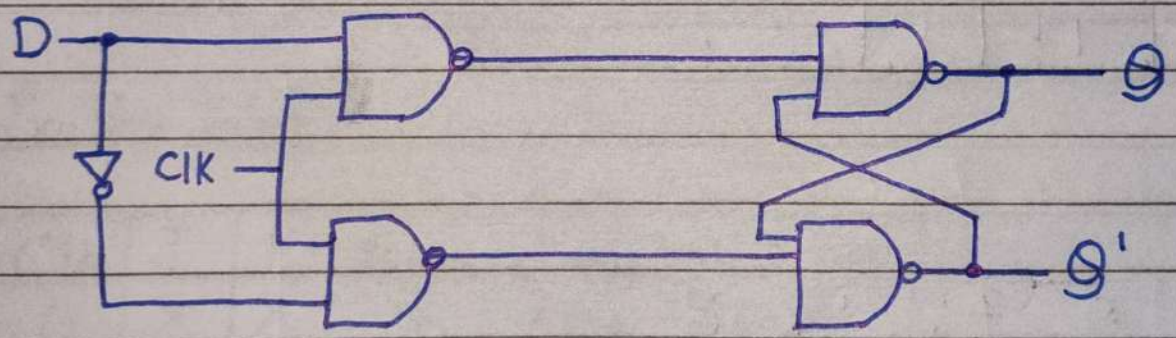
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D Flip Flop

Block Diagram



Logic Diagram (SR-to-D)



Truth Table

CLK	D	Q_{n+1}	State/Action
0	X	Q_n	No Change
1	0	0	Reset
1	1	1	Set

Characteristic Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Equation

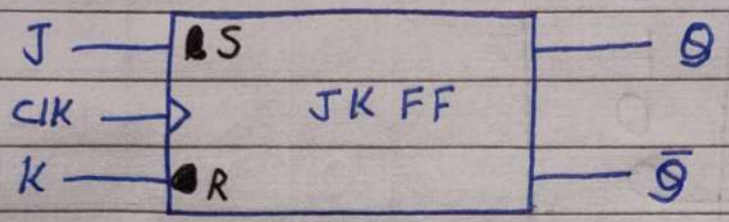
$$Q_{n+1} = D$$

Excitation Table

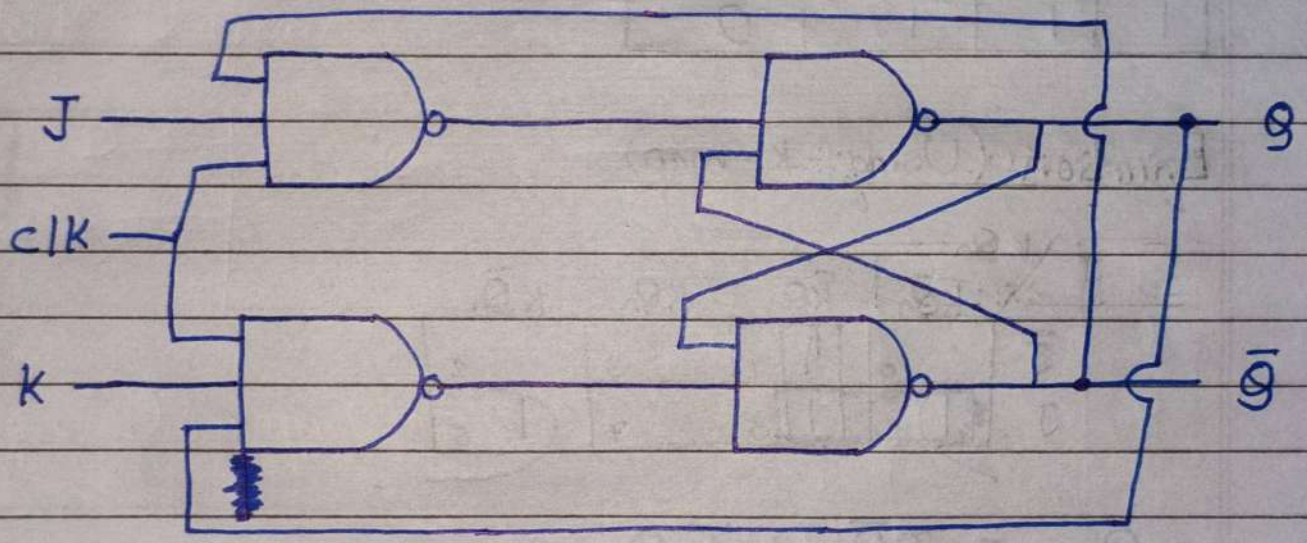
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

J-K Flip Flop

Block Diagram



Logic Diagram



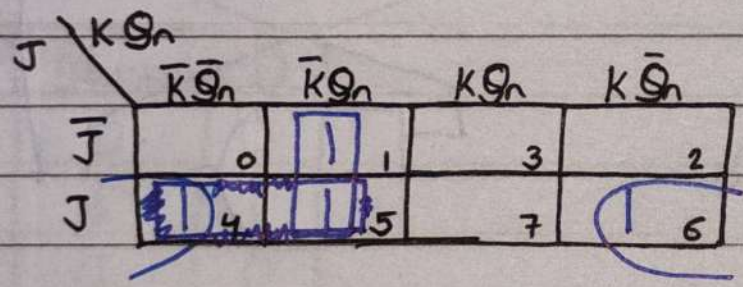
Truth Table

CLK	J	K	Q_{n+1}	State / Action
0	X	X	Q_n	No Change
1	0	0	Q_n	No change
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\bar{Q}_n$	Toggle

Characteristic Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Equation (Using K-map)



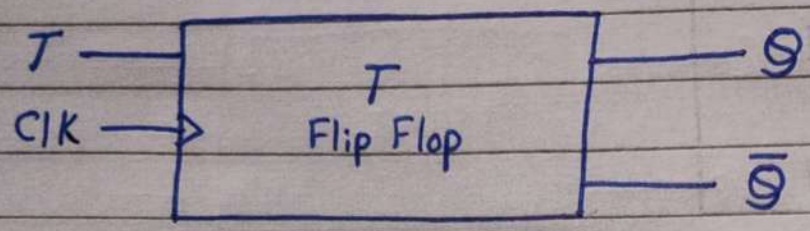
$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

Excitation Table

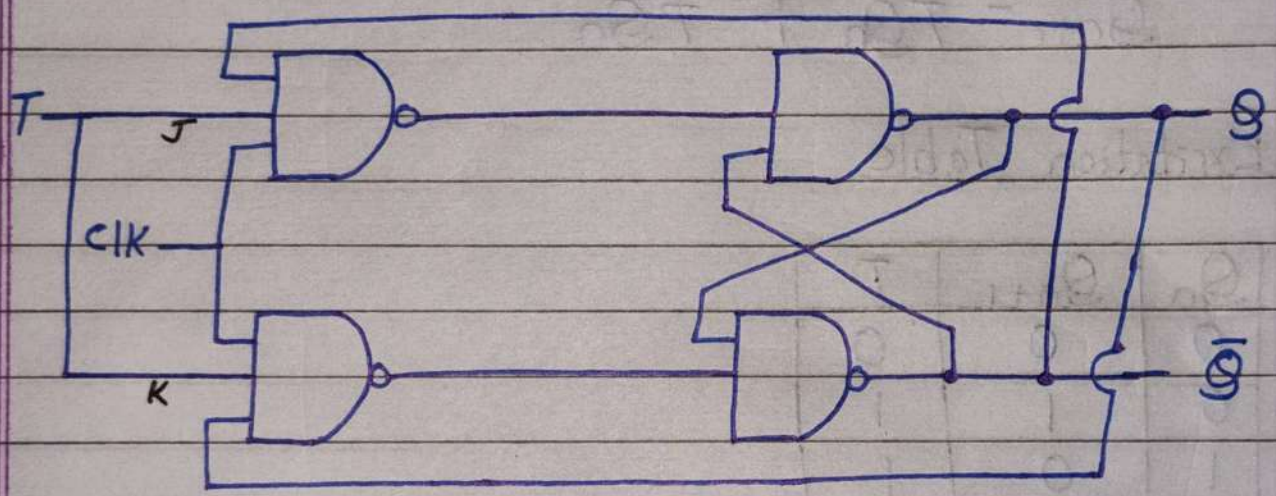
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

T Flip Flop

Block Diagram



Logic Diagram



Truth Table

CIK	T	Q_{n+1}	State/Action
0	X	Q_n	No Change
1	0	Q_n	No Change
1	1	$\bar{Q}_n$	Toggle

Characteristic Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Equation

$$Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n .$$

Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

More Topics Remaining